Hiring: Integration Challenge

Introduction

Our main goal for this exercise is to get an idea of how you are to work with and how you approach your engineering work. That insight is more important than your actual working solution at the end of this exercise.

Please record your screen and talk us through the coding exercise as you work through it. Don't hesitate to think out loud—that's the interesting part!

We expect this to take around an hour or two. If you exceed this window, please focus on the areas you deem most critical to the project's success and use comments to explain your architectural intent for any remaining parts. We value your engineering prioritization, trade-offs and thought process more than a 100% complete solution.

Please feel free to reach out to us at anytime if you need more information, or if something is unclear.

Overview

Tilla is a multi-tenanted platform. This means that a lot of different clients use the same platform and each have their own flavour of data that they provide us with. Some flavours have bad mappings, others have malformed strings, some others may completely deviate from the API schema that **they** provide.

As an integration-heavy platform, we must gracefully cater for all different flavours of data while ensuring every row of data that touches our database is sanitised, safe and free of errors without blending into the flavours from other tenants.

One of the ways we achieve this is by using a strict data contract for what fields we expect in our database.

Today, you will get a taste of this flavour with a fun new integration project given to you!

Goal

The goal of the assignment is to simulate a feature request assigned to you. The team has granted you a great degree of freedom in terms of how you approach the problem. You are expected to take ownership of this feature and ensure it meets our business requirements.

Your First Assignment

A New Integration Request From The Product Team

The product team has reached out to you with an exciting new integration! One of our clients has noticed that our seaport data is incomplete. An intern at their IT department has kindly made their seaport data available for us.
Please review the email thread forwarded to you by the product team:

> From: product@tilla
>
> Good day,
>
> Can you please take a look at the seaport data and see if we can use it to update our seaport data?
> Ideally, we want our seaport data to be kept in sync with what the client provides us.
>
> We anticipate that other clients might also provide us with seaport data in the future.
>
> The more data we have, the better!
>
> Cheers,
> Tilla Product Team
>
> > From: mike@shippingcompany
> >
> > Hi Tilla Team!
> >
> > I setup that connector you asked for to our seaports table.
> > The connector will automatically dump our seaport data to an xl file and upload it to
> >
> > an Azure Blob Storage. The automation runs ~2 times per day.

> You'll need to use this SAS token to connect:
>
> https://tillachallenge.blob.core.windows.net/challenge-data?sp=rl&st=2026-02-10T07:18:36Z&se=2026-04-01T15:33:36Z&spr=https&sv=2024-11-04&sr=c&sig=hWOx9eiybuxnOIIFwUqtNQF%2FMz5oyAwV8HXJWt6pYjM%3D
>
> Best regards,
>
> Mike the intern.
>
> Fracking Fast Freight Logistics

Engineering Guidance

After checking in with your mentor, they see this as a great opportunity to scaffold a new project, but advised keeping our current tech stack. We already have a seaports table in our main application. To ensure parity with the existing table, he has provided you with our data requirements sheet for seaport data:

Seaport Schema

Field Description	Field Name	Type	Requirement	Remarks
Port Name	`portName`	string	Mandatory	Name of the Port
Port Locode	`locode`	string	Mandatory	UN Locode
Latitude	`latitude`	float	Mandatory	
Longitude	`longitude`	float	Mandatory	
Timezone	`timezoneOlson`	string	Recommended	
Port Country ISO	`countryIso`	string (ISO 3166-1 alpha-2)	Nice To Have	

As part of the integration, they mentioned that the data should be made viewable on a simple dashboard for the support team to manually review the data for correctness.

Objectives

- Start recording
  - This is the first thing you should do.
  - Think out loud. We want to hear your thought process .
  - When making design decisions (eg. choosing the architecture approach for the ETL) vocalise your reasoning. Eg. “I am thinking approach X usually works well for ETLs, but might too heavy to run in Y environment so I will rather go for approach Z”.
- Scaffold a new project:
  - Scaffold a new full stack project
  - Create an integration (ETL) layer that syncs the data from the Azure Blob Storage
  - The data is loaded into a new database that you create.
    - Use Prisma for the ORM
    - Use a local running PostgreSQL database
  - Create a simple React app to fetch and render the seaport data
    - Use any React framework of your choosing (NextJS, CRA, Vite)
    - Fetch the data from the backend using GraphQL
- Upload the project to Github
  - See “Deliverables” below

Strict Requirements

- Give clear instructions in a README for our team on how to run your solution
- A script to bootstrap the app with minimal setup is advised
- The code must compile! If we cannot run your app we cannot properly assess how it performs

Business Insights

Valuable business insights that should guide your decision making:

- The platform consists of multiple clients.
- Other clients might provide us with seaport data in the future
- Data must be unique. Duplication is bad.
- Data quality is very important for our platform to run smoothly

Recommended Tech Stack

Here is a glimpse of the tech stack we love at Tilla

- Backend
  - NestJS
  - GraphQL
- Database
  - PostgreSQL
  - Prisma ORM
- Frontend
  - React
- Infrastructure
  - AWS
  - Serverless
  - CDK

While you have reasonable freedom to approach this as you like, we'd encourage you to use our tech stack to guide your decisions when scaffolding the project

Extra Questions

We'd love to hear your thoughts on some of these questions. Please don't spend more than a minute or two on each question.

- What are some edge cases you would take care of before shipping this to production?
- How would you scale this to handle high amounts of traffic?
- What's important for you to work well in a fully remote team?

Deliverables

- Invite @umartayyab, @Calvin-Tilla, @akshatamohanty, and @AleSua93 to a GitHub repo with your completed project
- A video of your screen recording (unlisted YouTube video, Loom, ... anything works)
- Answers to the questions above can be either in the video or written down in the README of your repo

AI Policy

We like AI and trust you to use it wisely: but we want your code, your thinking. Use AI to scaffold, speed up, debug, or explore ideas, but make your submissions production-ready and reflective of your own problem-solving style. In short: *trust, but verify*.